Socio-Economic Segmentation (County Level)

Date: 25 March 2026

Dataset Used: US Census Demographic Data (acs2015_county_data.csv)

Objective:

Build a machine learning model to segment U.S. counties into distinct socio-economic groups using clustering techniques based on income, employment, population, and poverty indicators.

Task Performed:

1. Imported the ACS 2015 county-level dataset from Kaggle containing demographic and economic data for over 3,000 counties.
2. Performed initial data exploration to understand dataset structure, variables, and distributions.
3. Cleaned the dataset by removing missing or inconsistent values to ensure accuracy.
4. Selected relevant socio-economic features such as total population, median income, poverty rate, unemployment rate, employment categories, and mean commute time.
5. Applied data standardization to normalize feature values and eliminate scale differences.
6. Used the Elbow Method to determine the optimal number of clusters for segmentation.
7. Implemented the K-Means clustering algorithm to group counties based on similar socio-economic characteristics.
8. Assigned cluster labels to each county and performed group-wise analysis.
9. Created visualizations (scatter plots, box plots) to compare clusters and identify patterns in income, poverty, and employment.

Conclusion:

The model successfully grouped U.S. counties into distinct socio-economic segments using indicators like income, poverty, and employment. It highlights clear differences between economically strong and weaker regions, revealing important regional patterns. These findings can support informed decision-making in planning and resource allocation, demonstrating the practical value of clustering in socio-economic analysis.